# 3MB3 Final Project

## Arms Race

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# Overview

**Scenario**: Countries p and g are locked in an arms race where each is:

- Encouraged to increase military spending based on each other's spending
- Discouraged to increase spending based on their own spending

**Model Overview**:

- $\Delta$spending = x(other country's spending) - y(own spending)
- x and y measure the severity of encouragement or discouragement to spend

**Research Question**: What happens to the long-term spending of each country?

# Arms Race Base Model

## Model Equations

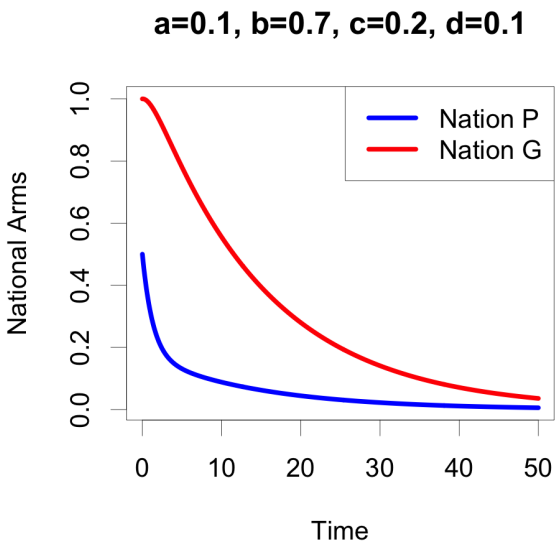
$$\frac{\partial p}{\partial t} = ag(t) - bp(t)$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t)$$

- Feasibility Condition: $(p(t), g(t) \geq 0)$
- $a, c$: Mutual Fear Factor $(a, c > 0)$
- $b, d$: Expenditure Factor $(b, d > 0)$

# Long Term Behavior

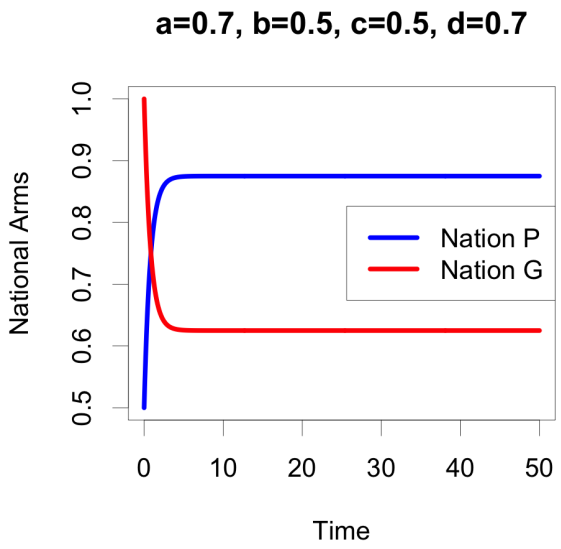
In the long run, we predict:

If: $bd > ac$



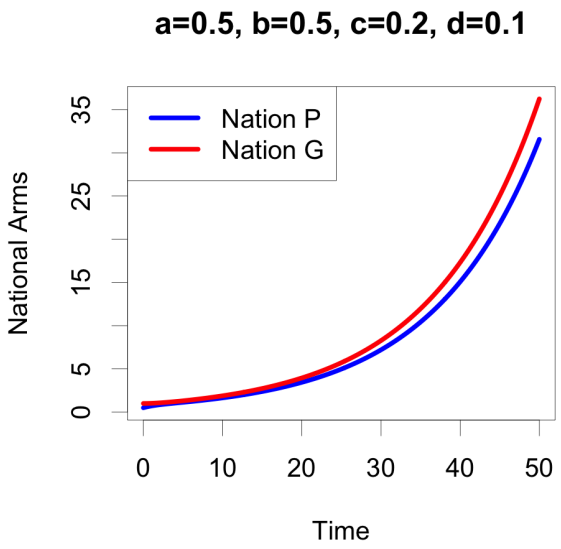

Complete mutual disarmament $(p^\infty, g^\infty) = (0, 0)$

If: $bd = ac$ **edge case**




Stabilized spending $(p^\infty, g^\infty) = (\frac{cp_0+bg_0}{a+c}, \frac{dp_0+ag_0}{a+c})$

If: $bd < ac$




Perpetual arms race $(p^\infty, g^\infty) = (\infty, \infty)$

# Grievances and Goodwill

Suppose countries p and g have underlying goodwill/grievances that influence their spending
The model equations are therefore:

$$\frac{\partial p}{\partial t} = ag(t) - bp(t) + r$$
$$\frac{\partial g}{\partial t} = cp(t) - dg(t) + s$$

- $(r, s) > 0$ for goodwill and $(r, s) < 0$ for grievance
- Provide further elaborating points

# Long Term Behavior

- provide plots
- provide stability conditions and convergences
- This model will take on negative values so discuss that by either capping the value or other methods

# Mutual Logistic Fear

**Limitation of Current Model**: Does not account for the limited resources each country has
**Solution**: Introduce logistic growth

$$\frac{\partial p}{\partial t} = a(1 - \frac{p(t)}{K_p})g(t) - bp(t) + r$$
$$\frac{\partial g}{\partial t} = c(1 - \frac{g(t)}{K_g})p(t) - dg(t) + s$$

- Provide further elaborating points
- Provide further elaborating points

# Long Term Behavior

- provide plots
- provide stability conditions
- provide convergences (if possible)

# Thank you for listening

[0] Presentation theme by Ricardo B. Lourenco